BioSpecInfo

Software Bill of Materials (SBOM)

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Versione **bsi-v229** · commit `ab91286`

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Software Bill of Materials (SBOM)

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Commit	`ab912865bb8e33f133c136c19bcd08fc31341282`
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Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer _{Daniel Probst}	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`
`three.min.js`	`9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2`
`three_bloom.js`	`42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090`
`gltf_loader.js`	`5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6`
`3Dmol-min.js`	`06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff`
`smiles-drawer.min.js`	`955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2`
`vendor/pdfjs/pdf.min.mjs`	`bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259`
`vendor/pdfjs/pdf.worker.min.mjs`	`145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6`
`lib/sql-wasm.js`	`77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f`
`lib/sql-wasm.wasm`	`438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed`
`lib/dimuon.js`	`61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d`

2. Code written for this project

File	Size	Role
`index.html`	4677 kB	Main application: interface, teaching sections, chemical data.
`bsi-ai-hub.js`	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
`bsi-spettri.js`	35 kB	IR/NMR spectral prediction engine over the molecular graph.
`bsi-cheminfo.js`	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
`bsi-pretsch.js`	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
`bsi-nmr.js`	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
`bsi-geom3d.js`	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
`bsi-fisica-comune.js`	15 kB	The shared home of the physics sections: header, tabs, sliders, canvas charts, language.
`bsi-fis-schrodinger.js`	22 kB	1D Schrödinger equation: finite differences, Sturm bisection, inverse iteration, Crank-Nicolson, Morse for real molecules.
`bsi-fis-hartreefock.js`	29 kB	Restricted Hartree-Fock in the STO-3G basis: McMurchie-Davidson integrals, Boys function, SCF with DIIS, molecular orbitals.
`bsi-fis-rotovib.js`	16 kB	IR and Raman rovibrational spectra of diatomics from molecular constants; combination differences.
`bsi-fis-termostat.js`	11 kB	Statistical thermodynamics: partition functions, entropy, heat capacity, Boltzmann populations.
`bsi-fis-dinamica.js`	12 kB	2D Lennard-Jones molecular dynamics: velocity Verlet, thermostat, virial, radial distribution.
`bsi-fis-matematica.js`	24 kB	Mathematics for chemistry: FFT of the FID, Hückel, Runge-Kutta for kinetics, least squares with errors.
`bsi-fis-cristalli.js`	20 kB	Crystallography: Bravais lattices, Miller indices, powder pattern with structure factor and absences.
`bsi-fis-nucleare.js`	14 kB	Radiochemistry: Bateman equations, radiometric dating, Weizsäcker semi-empirical formula.
`bsi-mat-cas.js`	63 kB	Computer-algebra engine: parser, exact fractions, canonical form, derivatives, verified antiderivatives, truncated series, limits.
`bsi-mat-basi.js`	26 kB	Mathematics basics: significant figures, logarithms and pH, equations with working, practice.

File	Size	Role
`bsi-mat-funzioni.js`	17 kB	Function analysis: domain, zeros, extrema, inflections, asymptotes; families and transformations.
`bsi-mat-calcolo.js`	21 kB	Derivatives with rules, integrals and Riemann sums, Taylor polynomials, limits.
`bsi-mat-lineare.js`	38 kB	Exact-fraction linear algebra: step-by-step Gauss-Jordan, determinant, inverse, eigenvalues, balancing as a null space.
`bsi-mat-ode.js`	21 kB	Differential equations: direction field, Euler/Heun/RK4, phase plane and equilibria from the Jacobian.
`bsi-mat-multi.js`	22 kB	Functions of several variables: surfaces, critical points from the Hessian, Maxwell relations, exact differentials.
`bsi-mat-complessi.js`	20 kB	Complex numbers (Argand plane, roots, Euler) and Fourier series.
`bsi-mat-risolutore.js`	104 kB	Mathematics exam solver: step-by-step working for 16 exercise types.
`bsi-mat-esame.js`	53 kB	Istituzioni di Matematiche I and II exam: syllabus, solver, solved papers, simulation.
`bsi-fis-esame.js`	97 kB	General Physics I and II exam: syllabus, 48 problem models, solved papers.
`bsi-chi-esame.js`	77 kB	Chemistry exam from basic to expert: 40 problem models, solved papers.
`bsi-fis-elettrodi.js`	14 kB	Electrode kinetics (Butler-Volmer, Tafel) and transition-state theory (Eyring).
`bsi-fis-superfici.js`	18 kB	Photophysics (Jablonski, Stern-Volmer), adsorption isotherms (Langmuir, Freundlich, BET), polymers.
`bsi-mat-statistica.js`	27 kB	Laboratory statistics: t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD and LOQ.
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
`bsi-lingue.js`	187 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.

File	Size	Role
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	6 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	194 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
`index.html`	`3cf6ef4cdc920a984d13bdd8e7ebf8c4c747c20cf924bf5484c2ac8e7d76d7b3`
`bsi-ai-hub.js`	`985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476`
`bsi-spettri.js`	`be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9`
`bsi-cheminfo.js`	`59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc`
`bsi-pretsch.js`	`27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3`
`bsi-nmr.js`	`881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72`
`bsi-geom3d.js`	`a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054`
`bsi-fisica-comune.js`	`092cde6b44f779694b0aa865f407a2e011a6d047878bf8e2fa3c819d6f3c23e0`
`bsi-fis-schrodinger.js`	`b032084511102c7231bcb244b5bb647128a080a18e9e6d706ec4c727ac29d17d`
`bsi-fis-hartreefock.js`	`8210d2a1501fe6e3f57c0ce8a0e6d5d78432274792833d50bc2e821fc7566fde`
`bsi-fis-rotovib.js`	`428182687a76111f013986127ee49beb7393498387b146bd0694ff16bea7d909`
`bsi-fis-termostat.js`	`0bb9f04d7d5b144791764fa4efd9f8e658be5b5d78b40f94e18752c2366589a9`
`bsi-fis-dinamica.js`	`f1d76c737ce7f75cdc9c75665194180a4cdf401e47a105583838f31c85c9b906`
`bsi-fis-matematica.js`	`616a2d4f338fe1d371f5ba3fd4fe569e94b2d94626a8d323bba8ae58f8550f62`
`bsi-fis-cristalli.js`	`f5283d63ff075ddc19f9d200194528366f38294909db5872e7604b46e0a7d7d5`
`bsi-fis-nucleare.js`	`8ae148a2f37a40f6902faa4148e44468b4d659b6c7d1230378de92c2703a5874`
`bsi-mat-cas.js`	`a5f79240bca5d63154f67ca30a01115dd15c3a93812438c7fcaea1665e3587ac`
`bsi-mat-basi.js`	`db30e16c58bde3638f13de5c3311a00598628676b7b418dc023b175efa77c214`
`bsi-mat-funzioni.js`	`68f8780d4fb4fcda6f028e4749adef333253efbf270d0caab47def5bdf30c66c`
`bsi-mat-calcolo.js`	`c3363448c4d773d29c4f680eddb79e9ad55a5eb87b033a728754df8cbee2a631`
`bsi-mat-lineare.js`	`da33ea18f2b14c21c6e630be4a1ccfe2e35e911489bdd4ededd60c0f533ac94f`
`bsi-mat-ode.js`	`698725d49dc2bf862161914e41d38c3fafd33069021eae3b11b4973c146b52ce`

File	SHA-256
bsi-mat-multi.js	1040861c3b9d340febe8c4a78d212b168eb42752ee22589facf8566a401234d7
bsi-mat-complessi.js	52a2ac5594a697d3a008d9fd629bee60fcff96ea76aa16c2ccda63eafd43f316
bsi-mat-risolutore.js	b888e18d26d91b78a392c224544e93edb2a660e506fe363d030d2c94083b7b27
bsi-mat-esame.js	e9d920b422d395e48a00a607af88879e6dc5a3646d7341f0a5677bc8f0e57af7
bsi-fis-esame.js	1f30f568364cd4b0a343d1befcf1c2d0e5233f0db9a98e7770fd1b734a1a5d46
bsi-chi-esame.js	85fe03cae65402bbc6cec099594d8c87be912694d0edcefce5d3cf518b476c35
bsi-fis-elettrodi.js	249bb92ac39b78e916eaa0ecf80a2b1aac5d10249bb6b51628f61839c4b64240
bsi-fis-superfici.js	3daec64d2c4755e70279b8ef2f433e4c98d6e90ff7d8c7d9c0689a24f080abd8
bsi-mat-statistica.js	0b0c82afa13173b84308589cf16fbaa8b30c5e575fd8fdee00034b70c34d17fc
bsi-nmr2d.js	a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78
bsi-moldraw.js	45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e
bsi-documento.js	0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9
bsi-quesito.js	6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf
bsi-spettrolettore.js	14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256
bsi-mol3d.js	93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893
bsi-lingue.js	ba192b53f95ff6ae1dd75b0169d8a7820da42f6d110286a52dd2af1e3dbecac5
bsi-molingue.js	e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb
bsi-pannelli-lingua.js	4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93
sw.js	88adcc4f934cdb515d59c23dbca1c07e3b1e19738808ff6349881330709e2f6c
rdkit_lab.html	066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad
astro.html	3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d
chimorga.html	047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d
proxy/spectra-proxy.js	808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
playwright-core	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

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5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

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